
Inductive Bias Meta-Learning with Generative Models

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Abstract

This paper investigates inductive bias in machine learning models. By inductive bias we mean the preference of a model for certain types of functions or data structures over others. To analyze the inductive bias of a fixed model, we consider a problem of finding data that this model can fit and generalize on particularly well. Previous work demonstrated that generating labels for a fixed dataset allows one to extract the inductive bias. Here, we extend this approach by proposing a method for generating full synthetic datasets. We train a generative model to produce datasets on which the target model achieves strong generalization performance. We test the proposed framework on CNN and RNN, and analyze obtained datasets for each model.

Keywords: inductive bias, meta-learning, generative models.

1 Introduction

TODO

Contributions. Our contributions can be summarized as follows:

- We present...
- We demonstrate the validity of our theoretical results through empirical studies...
- We highlight the implications of our findings for...

Outline. The rest of the paper is organized as follows...

2 Related Work

Topic #1. TODO

Topic #2. TODO

3 Preliminaries

3.1 General notation

In this section, we introduce the general notation used in the rest of the paper and the basic assumptions.

3.2 Assumptions

TODO

4 Method

5 Experiments

To verify the theoretical estimates obtained, we conducted a detailed empirical study...

6 Discussion

TODO

7 Conclusion

TODO

References

A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO